Evolution of Electronic Warfare History

112-BOLCEW01

Learning Objective

During this lesson we will:

Discuss the evolution of Electronic Warfare (EW) and explain how EW fits into the Army Cyber Branch

Learning Step Activities

Electronic Warfare Evolution:

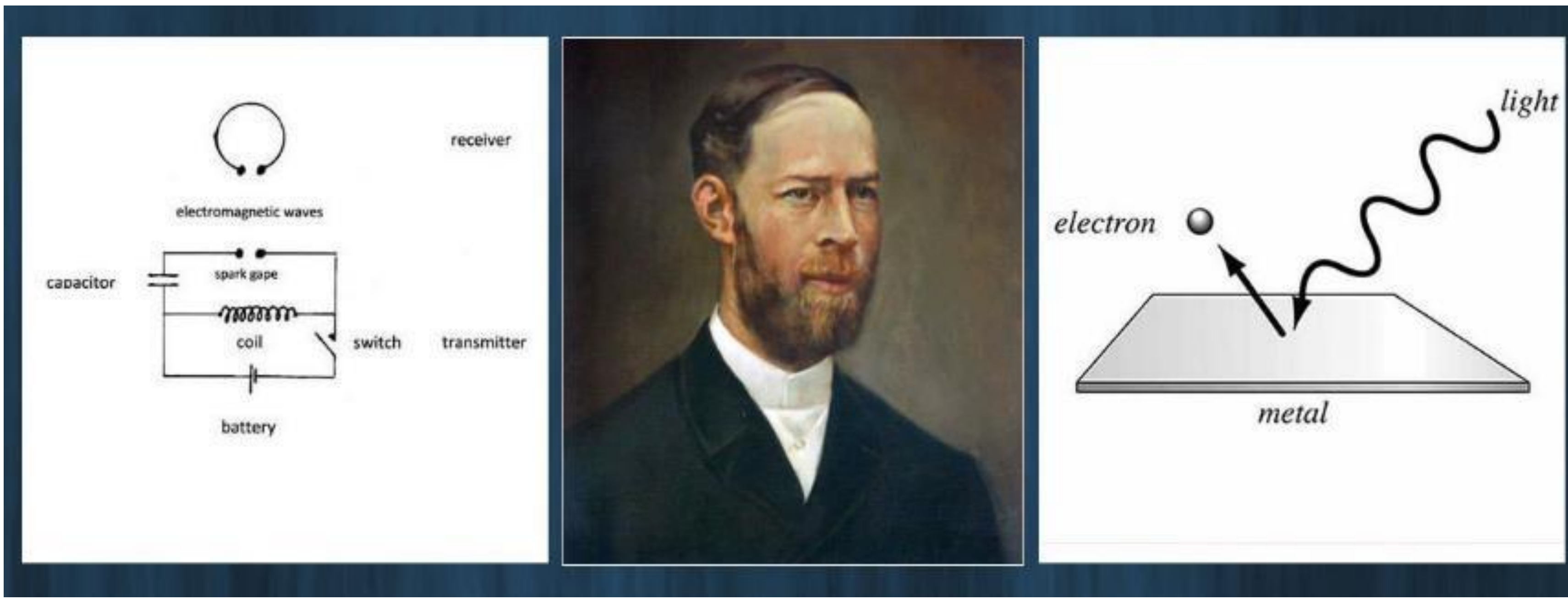
- Significant Pre-World War I Electronic Warfare Events
- Significant Events for Electronic Warfare during the 20th Century
- Significant Events for Electronic Warfare during the 21st Century
- Current/Recent Conflicts for Electronic Warfare
- Emerging EW Technologies
- EW's Organizational Journey within the Army 1945-Present

Pre-World War I Electronic Warfare Events

Prominent Figure in Radio Frequency History

1888:

- Heinrich Hertz demonstrated electric sparks propagate into space
- Was considered a “scientific phenomenon,” not for its potential as a communication method



Heinrich Hertz

Break-Through Year: 1897

1897:

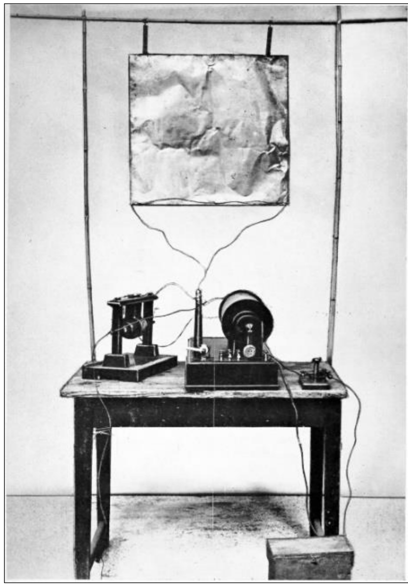
- Based Hertz's study of radio waves, **Guglielmo Marconi** invented the first radio

- Sent the first ever wireless communication over open sea

    - Between the Bristol Channel to Lavernock Point (South Wales) to Flat Holm Island totaling 3.7 miles

- Marconi's radio patent was approved and production begins

Guglielmo Marconi

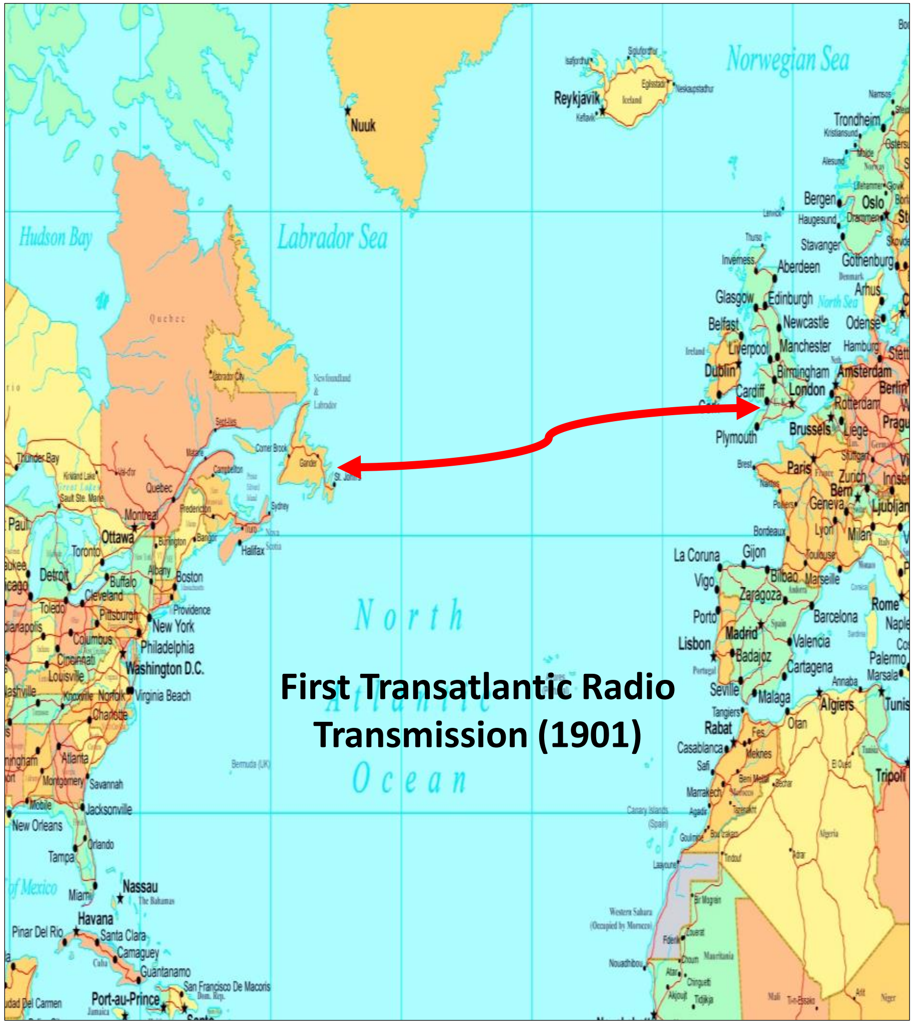


Marconi's First Transmitter w/ Monopole Antenna

Marconi Wireless Milestones

1899-1903:

- 1899: Greater distances of radio transmission was achieved by 30 miles near Dover, England
- 1901: Marconi succeeded in sending a radio transmission **over 2000 miles** between England and Canada
- 1903: Marconi comes to the U.S. and helps President Roosevelt send first wireless message to the King of England



Pre-World War I Electronic Warfare Events

1901: First Deliberate Radio Jamming

America's Cup Yacht Race and Jamming:

First ever occurrence of DELIBERATE radio jamming:

- 1901 America's Cup Yacht Race between The Columbia and the Shamrock II
- Wireless telegraph companies hired by newspapers to send reports
- First newspaper to get results would sell more than competitors
- Transmitter held down over an hour purposely jammed all other wireless news updates

Pre-World War I Electronic Warfare Events

1903: Marconi is Victim of Electronic Attack

***June 1903*:**

- Marconi organizes demonstration at London's Royal Institution
- Wants to prove wireless transmission using his apparatus was <u>safe from interference or spying</u>

Before Marconi transmitted message:

Operator heard unexpected transmission in Morse code – "There was a young fellow of Italy, who diddled the public quite prettily…"

Marconi's "safe" wireless transmission was the victim of Electronic Attack!

Nevil Maskelyne
"Scientific Hooligan"

Pre-World War I Electronic Warfare Events

1903: Navy attempted to incorporate jamming into fleet maneuvers & divided a force into two groups – Blue and White

BLUE **VS** **WHITE**

- Blue Force mission was to engage the White Force before it reached the New England coast
- Four "Blue" force ships with radio were positioned as a screen to warn of the White approach
- They would radio the USN radio stations on the coast upon discovery of the White force
- Upon spotting the White force, they commenced their wireless message

- White force, starting 500 miles east of Cape Cod, was to attempt to land troops on the coast
- Only one of the White ships had a radio – the USS Texas
- The USS Texas planned to jam the Blue Force sighting reports as soon as they heard them begin transmitting
- After the first three letters came through the USS Texas radio operator started to jam, but his officer stopped him because he wanted to hear the whole message…

- When the officer told him to go ahead and jam, he explained it was too late – the message had already gone out…
- **One of the First Recorded instances of Conflict Over Intelligence Gain-Loss (IGL)**

Pre-World War I Electronic Warfare Events

1906: Army Signal Corps' First Wireless Platoon

- Army Signal Corps began utilizing radio by 1899
- The Signal Corps' chief electrical expert, 1LT George Squier traveled to London to study under Marconi
- 1906: Signal Corps had its first wireless platoon
  - Established communications from Cuba to Key West during a peacekeeping operation

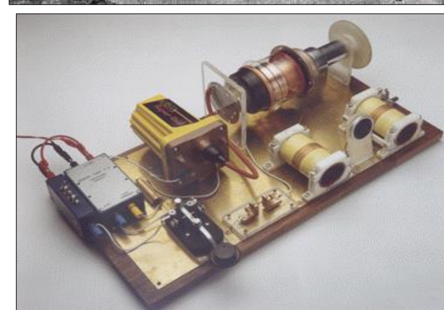
Early Army Signal Corps w/ Radio

Pre-World War I Electronic Warfare Events

Combat Jamming: Russo-Japanese War 1904-1905

Russia overheard Japanese radio transmissions

- Understood the importance
- Russian Radio Operators used "Spark" transmitters to jam Japanese transmissions
- Proof: Any transmitter is a jammer
- **<u>1904: First documented use of EW in Combat</u>**
- Russians commemorate this incident on 15 April each year as a "Day of Radio electronic Warfare"



Spark-Gap Transmitter

CHECK ON LEARNING

Q: What famous incident saw the first ever occurrence of deliberate radio jamming?

A: 1901, the America's Cup Yacht race when a wireless telegraph company intentionally blocked the communications of their competitors to get their news story out first.

Q: What conflict saw the first use of deliberate jamming in combat?

A: The Russo-Japanese war of 1904 when Russian radio operators used their Spark transmitters.

EW Events during World War I

World War I starts in August 1914

- Assassination of Archduke Ferdinand of Austria
- World at War with Germany

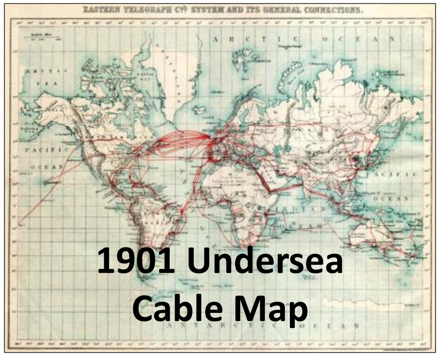
Information Warfare begins:

- British Cable Ship (CS) Alert cuts German undersea telegraph cables
- Forcing Germany to use the Radio in the air
- Germans report the British are attempting to jam wireless communications

Results:

- Germany invested heavily in powerful wireless stations, even operating a large station on Long Island while the U.S. was still neutral

British CS *Alert*



1901 Undersea Cable Map

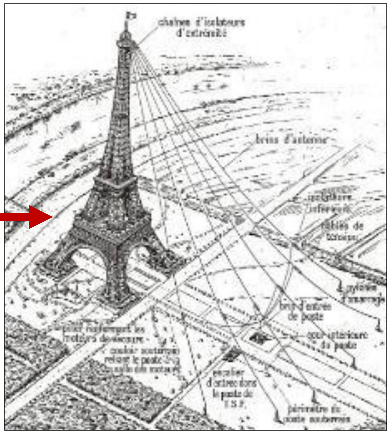
German Owned Station on Long Island

EW Events during World War I

WWI: Birth of modern Signals Intelligence (SIGINT)

Both Allies and Central Powers used wireless

- Signals were impossible to hide from enemy
- Listening stations set up throughout the theater
- Example: Eiffel Tower was one



Encryption becomes necessary

- Russians began broadcasting at night thinking Germans were asleep
- U.S. Army Signal Corps began making cipher devices

Results:

- Cryptoanalysis and codebreaking became important
- SIGINT is born

WWI Radio SIGINT

EW Events during World War I

WWI: Birth of modern Signals Intelligence (SIGINT)

Army's Cipher Bureau Military Intelligence-8 (MI-8)

- Established by Mr. Herbert Yardley
- Army's Signal Corps worked closely with MI-8
- Passed data collected for transcription and analysis for enemy operations and intentions

Herbert Yardley

1915: U.K.'s Royal Navy develops Direction Finding (DF)

- Stations set along the English Coast
- Find bearings on any unit operating in the North Sea

English Coastal DF Station

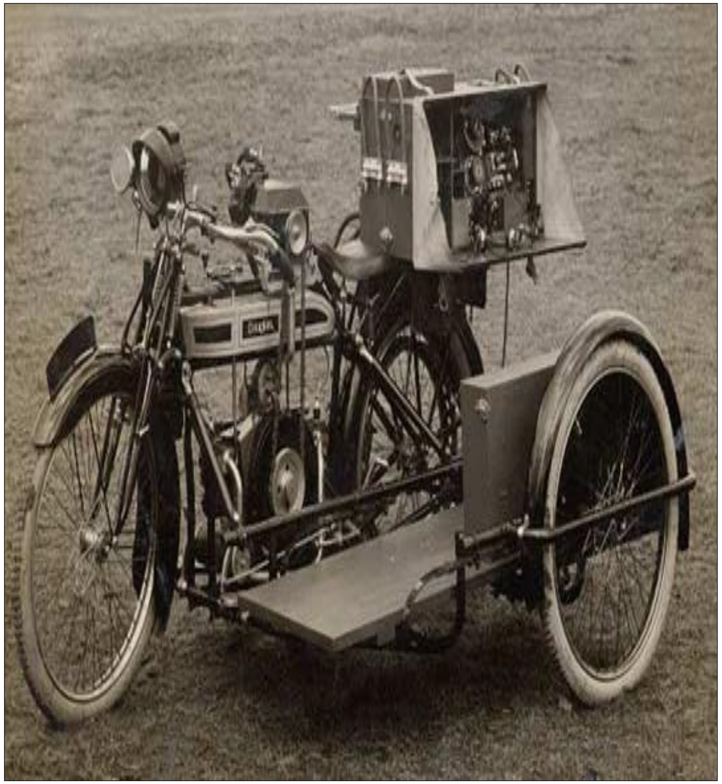
WWI: Advancement of Mobile Wireless

Wireless communications became mobile

- At start of WWI, the technology was restricted to large permanent land sites and ships
- As the war went on, wireless became mounted on vehicles of the day
- Tanks, early aircraft, motorcycles, trucks

U.S. Navy breakthrough

- Shipboard wireless Direction Finding against submarines
- Allies able to track the Germans in all theaters



Motorcycle Mounted Marconi Radio

EW Events during World War I

UNCLASSIFIED

WWI: First use of Electronic Counter Countermeasures

Aircraft and blimps passing reconnaissance feedback:

- Both sides attempting to jam air-to-ground-reports
- Friendly machines also caused interference

Electronic counter-countermeasures introduced:

- Clapper-break on aircraft reduces effect of deliberate jamming
- Clapper-break also altered the pitch/tone of wireless message, helping to reduce interference

Aircraft Wireless Transmitter

Aircraft Receiver

UNCLASSIFIED

CHECK ON LEARNING

Q: During WWI, the encryption of data and the building of listening stations gave rise to what type of Intelligence?

A: Signals Intelligence (SIGINT)

Q: The creation of the “Clapper-break” for aircraft was the first use of what Electronic Warfare Technique?

A: Electronic Counter-Countermeasures

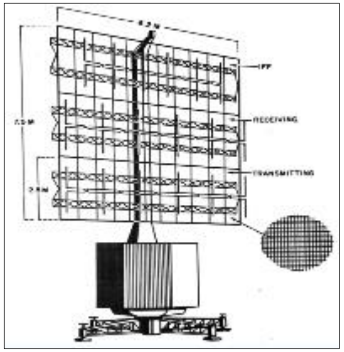
EW Events during the Inter-War Period

Radio Detection And Ranging = RADAR

Robert Watson-Watt

RADAR Technology Advances in Pre-WWII Years

- 1935: British scientist Robert Watson-Watt successfully bounced radio wave off an aircraft
- Proved the technology worked
- Early RADAR able to detect aircraft 17 miles away
- 1936: British RADAR range improved to 75 miles
- German’s also working with RADAR - detected aircraft at 12 miles



German Freya Radar System

EW Events during the Inter-War Period

RADAR, EW and birth of SEAD 1936-1939

U.K. leads the way with Early Warning RADARs

- Used to track aircraft
- “Chain Home” System used in defense

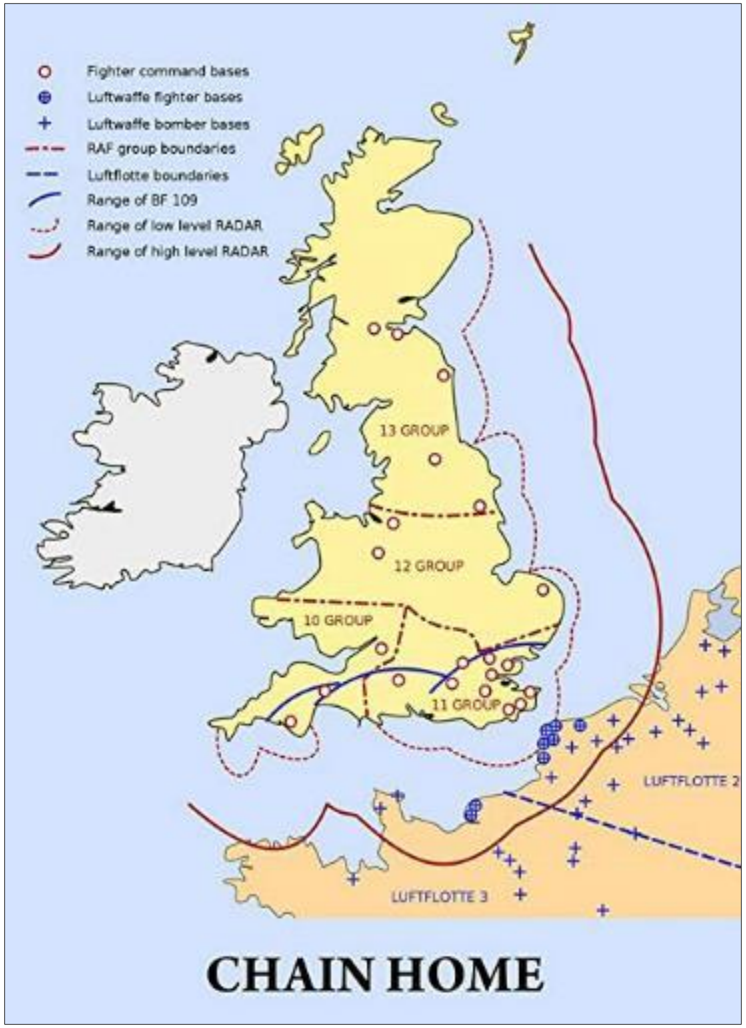
1938:

- First airborne jamming of RADAR conducted by Germans
- Various anti-jamming systems built into Chain Home system

1939: U.K. has 21 Chain Home stations

1941:

- US started organization devoted solely to development of radio countermeasures - the Radio Research Laboratory (RRL)
- RRL began working in earnest after German success

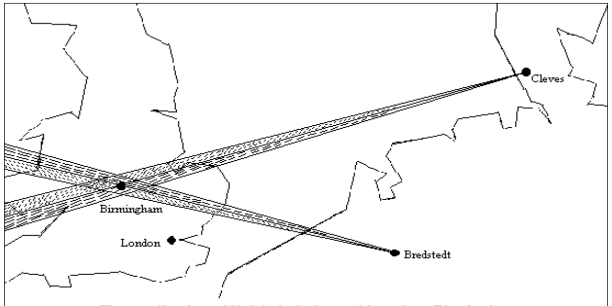


EW Events during World War II

Evolving TTPs

1940-41:

- Germany conducts large scale air campaign against the U.K. at night
- Germany used system called “Knickebein”
  - Ground transmitter radiating a beam to mark path for aircraft
  - Made night bombing accurate
- Great Britain Counters:
  - Discovered the system via airborne ELINT mission
  - Began jamming; reduced effectiveness



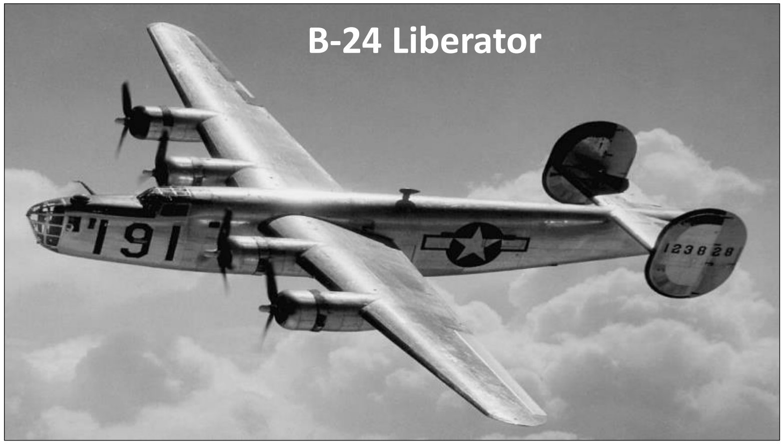
The application of Knickebein in a raid against Birmingham

EW Events during World War II

EW from the Air

1943: Army Air Force was turning some heavy bombers into ELINT capable aircraft

- Called Ferret Flights
- Outfitted aircraft used radar receivers to detect enemy radar on the ground as well as aircraft
- Done at night
- Used headphones to determine the strength of the signal received allowing pilots to gauge the angle of approach
- Bombers outfitted with RADAR jammers



EW Events during World War II

Electronic Attack on D Day

Mid-1942 - foil strips dropped in significant numbers by aircraft resulted in deceptive echoes on radar screens

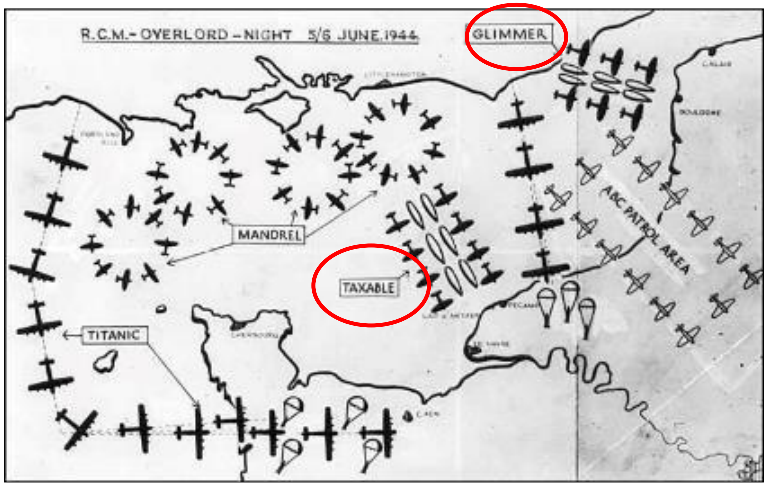
- U.S. called “Chaff” and the British referred to it as “Window”

5 June 1944: Operations TAXABLE and GLIMMER got underway

- Allied bombers flew carefully planned formations dropping form of chaff called “Rope”
- Created appearance on German RADARs of a fleet of ships headed for the French coast, **east of real invasion area**

6 Jun 1944 D-Day:

- Fleet of warships and landing craft approached the beaches of Normandy
- Switched on powerful jamming transmitters
- “Blinded the German defenders”

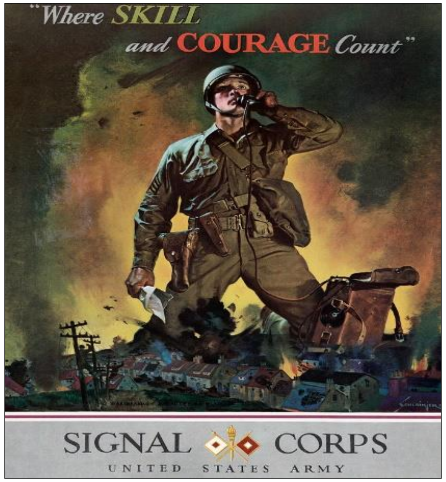


The Various D-Day Deception Schemes

EW Events during World War II

Army Signal Security Agency

Army's relative lack of jamming in WWII stemmed from fundamental differences in the way battles are fought on land



Army Signal Security Agency conducted radio countermeasures and radar deception through a unit called the Protective Security Branch

Army ground forces did not attempt communications jamming on a large scale because it would interfere with the vital flow of communications intelligence – the concept of Intelligence Gain/Loss

CHECK ON LEARNING

Q: What does RADAR stand for?

A: RAdio Detection And Ranging

Q: How was EW used in support of D Day?

A: Electronic attack in the form of jamming and chaff helped confuse German defenders and degrade their decision making ability.

Korean War 1950-1953

After WWII, Soviet Union occupied North Korea and the US assisted the South

Soviets and China supported the communist North

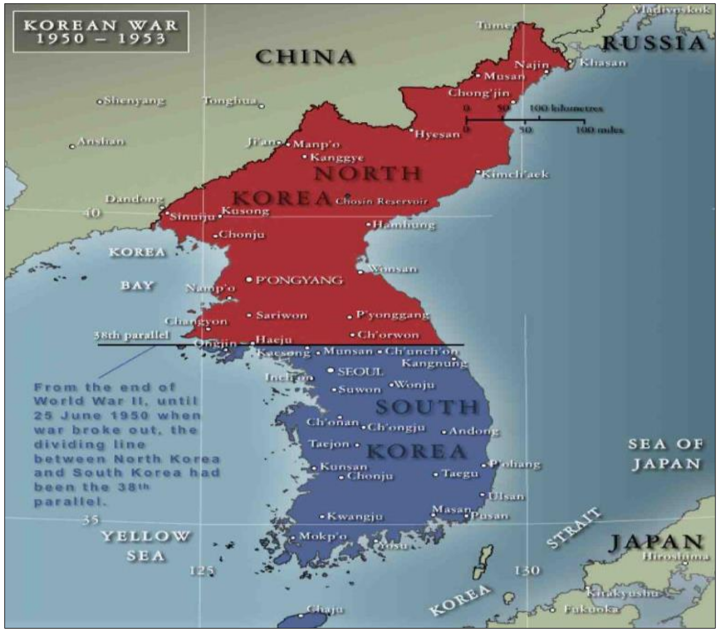
- 1950: North Korean forces crossed over the border and hostilities began

NATO forces (predominantly US) came to the aid of the South

EW research during Korean War:

- Countering Soviet made air-to-air and surface-to-air missiles
- US Ferret missions sought out North Korean radar locations
- US ELINT aircraft was the massive B-36 "Peacemaker"

WWII Trained personnel were recalled to help train the new EW crews



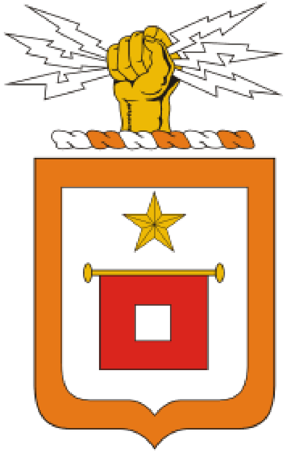
EW Events during the Onset of the Cold War

EW in the Army (1950s)

In 1954, the Army Signal Corps formed two Communications Electronic Warfare Battalions - the 72^{nd} and 73^{rd} at Fort Huachuca, AZ

- 1955: Operation SAGEBRUSH training exercise in Louisiana
- The 73rd CEW Battalion participated, and their jamming was so effective they were asked to leave

73d CEWB



Signal Corps

EW Developments in the Vietnam Era (1961- 1975)

1961: Ground EW/SIGINT in Vietnam

May 1961 ASA enters Vietnam to support South Vietnam Army through training and radio direction finding

Early ASA medium and long-range direction finding is underwhelming due to the **restrictive terrain and harsh environment**

ASA embeds **short-range direction-finding** teams with South Vietnam Army with **greater results but a higher level of risk**

Spec-4 James Davis of the Army Security Agency's 3rd Radio Research Unit is **ambushed and killed** after returning from a DF mission

Davis is sometimes referred to as the "First U.S. Serviceman Killed in Combat in Vietnam"

EW technicians like Davis were among the first US military personnel in hostile combat action in Vietnam

Army Security Agency

Davis with DF Equipment in Vietnam (1961)

SEAD Doctrine Comes of Age

North Vietnamese used radar and Soviet made SA-2 surface to air missiles

DoD became concerned about scale of losses, so new equipment were devised

By using a combination of Jamming decoys, Air units were able to suppress, fix and destroy AD systems. This tactic is known as **Suppression of Enemy Air Defense (SEAD)**

Douglas EB-66

- ELINT and Stand off Jamming aircraft

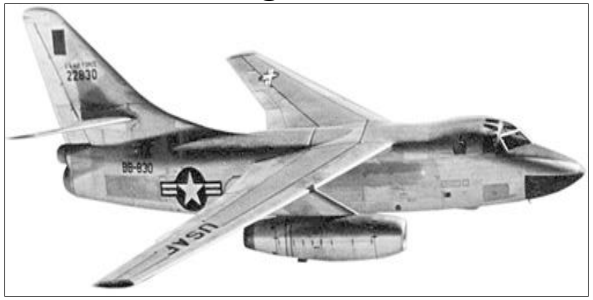
EA-6A Intruder

- Radar jammer

The F-100 and later F-105 Thunderchief fighters were code named **“Wild Weasels”**

- Worked as decoys, trying to get SAMs to fire
- Lock on ground RADAR location sites and destroy them
- Units also provided cover to fighter-bombers attacking conventional target

Douglas EB-66



F-105 Thunderchief

EA-6A Intruder

CHECK ON LEARNING

Q: Who is sometimes referred to as the "First U.S. Serviceman Killed in Combat in Vietnam"?

A: Spec-4 James Davis

Q: What does SEAD stand for?

A: Suppression of Enemy Air Defense

1982: First Lebanon War

Background – In the 1973 Arab-Israeli War against Syria and Egypt, Israel lost numerous aircraft to the Soviet designed SA-6 system

1982: First Lebanon War that pitted Israel vs. Syria and the PLO

- Days after the start of the conflict, Israel launched attack code named Operation Mole Cricket 19 in the Bekaa Valley where the strongest Syrian air defense system resided
- Operation was a suppression of enemy air defenses (SEAD) campaign launched by the Israeli Air Force (IAF) against Syrian targets

- Pre-battle ELINT gave Israel location of enemy air defense systems
- Remote piloted vehicles (RPV) fooled Syrians into launches from their SAM sites
- First time in history Western-equipped air force destroyed Soviet-built SAM network

Israeli F-15

Israeli Scout UAV

EW Developments in the Early 1990s

1990: Operation Desert Shield

RC-135 ELINT Capable

1990:

- Iraq invaded and occupied Kuwait
- U.S. deployed forces to Saudi Arabia and began coalition building with other nations

1991:

- Ground War begins supported by massive SEAD operation
- Victory achieved in 96Hrs

US uses lessons learned from Vietnam and Bekka valley to attack Iraq's IADS and achieve air FOM for air operations leading to total air supremacy

Spoof and Punch:

- US and Collation utilize ELINT aircraft to "map out" IADS
- Drones, Decoys and Jammers suppress and/or confuse IADs systems
- Stealth aircraft under cover of darkness destroy key early warning and C2 nodes
- Strike fighters followed by escort radar/comms Jammers destroy the bulk of SAM sites
- Remaining IADS are disjointed/confused and unable to repel attacks

EW Developments in the Early 1990s

Operation Desert Storm – Total Dominance

Iraq never recovered from "Spoof and Punch" operation the first night of the war

US destroyed the fiber optic cable switching stations, forcing the Iraqis to use microwave signals

Coalition forces owned the night, the skies over 10,000 feet, & the radio frequency spectrum

In the weeks that followed, the Iraqis were unable to do anything to change this

After Desert Storm ended in 1991, the US military wouldn't see much real combat until the Kosovo Conflict

The air campaign only saw two US aircraft lost out of 900 aircraft flying 38,000 sorties, a very impressive figure that is directly related to the various EW systems utilized

EW Developments in the Early 1990s

Land Battle: Desert Storm

Ground EW was overall Underwhelming

Tactical formations employed Combat Electronic Warfare/Intelligence companies (CEWI) to limited effectiveness

- Operation moved to quick
- Terrain in Iraq favored space based and arial assets that could "see" much larger areas in real time

Some Army and Marine units deployed TLQ-17A communications jamming equipment:

- U.S. leaders permitted jamming against high-level digitally encrypted signals, forcing the Iraqis to transmit in the clear
- These transmissions were left un-jammed for intel collection purposes

TLQ-17A Jammer

Army EW Begins to Change After 9/11

The terrorist attacks on September 11, 2001, led to:

- Operation Enduring Freedom (OEF) in Afghanistan - October 2001
- Operation Iraqi Freedom (OIF) - March 2003

Remote Control Improvised Explosive devices quickly became single most effective weapon against US forces

- By May 2005, over 7,700 casualties occurred from RCIEDs
- Army leadership looked to revive its own EW capability
- Counter RCIED equipment (CREW) was developed and put under control of the new Functional Area 29 (EW) specialty

These conflicts resulted in:

- **A change from LSCO to COIN centric Army**
- **Renewed need for EW in the Army Due to the rise in RCIEDs**
- **EW being used primarily as force protection focusing mainly on EA**

2014 Russian Involvement In Ukraine

Multi Domain Operations Take Center Stage

Following a revolution that ousted the pro-Russian government in Crimea, a series of increasingly hostile events led to the annexation of Crimea by Russia and a proxy war in eastern Ukraine by pro-Russian forces

Russia utilized CEMA and Information warfare:

- Russia utilized disinformation campaigns to mask their involvement and control the narrative/support
- Cyber attacks aimed at financial, industrial, and military targets significantly hindered Ukrainian opposition
- Russian EW forces utilized jamming, deception, and direction finding to fix and finish Ukrainian units

Ukraine was unprepared:

- Units lacked modern equipment and training combat CEMA
- Command was centralized and prone to "decapitation"
- Networks were not hardened, and EMS discipline was low

2020 2nd Nagorno-Karabakh War

Prelude of Things to Come

AUG-NOV 2020: The 2nd Nagorno-Karabakh War was fought by Azerbaijan against Armenia over a historically disputed mountainous region between the two nation that was controlled by pro Armenian forces.

Outcome:

- Clear Azerbaijan victory
- Capture of the 2nd Largest city in the region
- 6500-8500 KIA, 300+ tanks destroyed

20th Century Doctrine vs 21st Century Tactics:

- Azerbijan utilized drones to max effectiveness in both ISR and lethal strikes
- “loiter munitions” taxed a heavy toll on Armenia
- Armenia relied upon older and more static equipment making them easy targets to ES, Artillery, and Drones
- GNSS Jamming was extremely prevalent
- Decoys were used to expose and destroy Armenia IADS

2022-Present Russo-Ukraine War

LSCO In a Contested EMS

FEB 2022: Russia invaded Ukraine with close to 200K (including more than 80% of it's ground EW) soldiers attempting to capture the capital of Kiv IOT topple the government and incorporate portions of Ukraine into Russia

Current State:

- Russia failed to capture Kyiv
- Limited gains by Russia in eastern and southern Ukraine
- Bloody stalemate often resulting in trench warfare
- 300K-500K KIA from both sides (including 17 Russian generals)

Lessons Learned EW:

- Small UAS warfare is prevalent
- EW has been the most effective way to combat small UAS
- Defensive EA plays a large role against precision guided munitions
- Static Command Posts are easy targets in the EMS
- Russia overwhelming EW forces where not able to gain an advantage over Ukraine defenders
- GNSS frequencies are often denied or degraded
- Spectrum management and EMS reprograming is vital

Emerging EW Technologies

With the wars and conflicts expanding around the globe, new EW weapons began to be developed:

- Directed energy weapons focus energy beams to neutralize enemy forces
- Artificial Intelligence (AI) started being used to enhance spectrum management and Electromagnetic Support
- Counter-Unmanned Aircraft Systems (i.e., drones) are being used to conduct precision bombing attacks
- Joint All Domain Command and Control (JADC2) and Mosiac Warfare will change how DOD communicates and utilizes the EMS

Emerging EW Technologies

Destructive EA Weapons

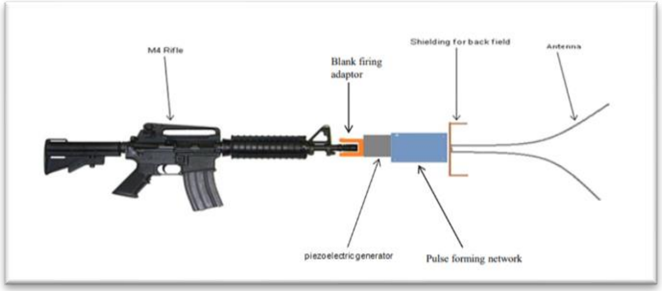
Counter-electronics High-powered Microwave Advanced Missile Project (CHAMP):

- Pinpoints buildings and knocks outs electrical grids
- Five units were fielded to the Air Force and tested in Utah on 7 targets in 2012

Laser Weapon System (LAWS):

- Navy used on Persian Gulf vessel in 2015

EMP Rifles:



- Army approved a patent in 2019 to turn M14 rifles into EMP guns allowing ground soldiers to destroy enemy electronics such as drones

Emerging EW Problem Sets

Defensive EA and Reprograming

Drones:
While not new technology, drones are being upgraded and becoming increasingly harder to stop through kinetic means

Precision Guided Munitions:
EW has been fighting these since Vietnam, however these munitions are now being used more at the tactical echelon where EW equipment is less common

Reprograming:
As Software defined radios become more prevalent, the ability for units to reprogram EW assets to match the threat is critical

EW's Organizational Journey within the Army
1945-2024

Army EW Organization in the 1940s and 1950s

- 1945 – Signal Security Agency separated from the Signal Corps and reorganized as the Army Security Agency (ASA) - placed under the leadership of the Director of Military Intelligence

- 1955 – ASA assumed control of Electronic Intelligence (ELINT) and communications Electronic Countermeasures (ECM) from the Signal Corps

Signal Corps

Army Security Agency

Army EW Organization in the 1960s and 1970s

ASA

INSCOM

- 1962 – ASA acquired the non-communications ECM mission from the Signal Corps

- 1976 – A new type of unit emerged: the Combat Electronic Warfare Intelligence, or CEWI Battalion
  - The first was the 522d MI (CEWI) Battalion at Fort Hood, TX, assigned to the 2d Armored Division

- 1977 – ASA was redesignated the Army Intelligence and Security Command (INSCOM)

522d MI CEWI Battalion

Army EW Organization in the 1980s and 1990s

- 1989 – After Berlin Wall came down, EW became less of a priority within the Army
- 1991 – Army CEWI units operated in Desert Storm in 1991 in aircraft and land operations, but this was the last main utilization for some time
- 1990s – With the end of the Cold War, the Army's EW capability slowly eroded and became ineffective
- **Military Intelligence community trained its last dedicated EW Officer in the 1990s**
  - **Leadership accepted risk**
  - **No replacement for aging electronic warfare/electronic attack systems**
  - **Army's participation in stability operations in 1990s did not require EW support**

EW's Organizational Journey within the 1945-2019

Global War on Terror and the New Army EW Career Field & School

FIRES Center of Excellence

After the 9/11 the Army moved to COIN operations in Iraq and Afghanistan. During this time RCIEDs caused a resurgence in EW within the army

- 2003 – Combined Arms Center (CAC) designated proponent for EW
- 2004 – Fires Center of Excellence, Ft. Sill, OK designated new EW School location
  - Focusing on Electronic Attack (EA)
- 2006 – First pilot course for Army Operational EW is held
- 2009 – Army approved 29 series career management field
  - **Functional Area (FA) 29 – EW Officer**
  - **290A – EW Technician (Warrant Officer)**
  - **29E – EW Specialist (NCO)**

Electronic Warfare Insignia

Approved – 26 October 2011

The Electronic Warfare insignia worn on the collar of the Army Service Uniform

Warrant Officer

NCO

EW's Organizational Journey within the 1945-2019

EW Transition to Cyber (2014-2018)

Jan 2014: Army announced transition of EW functional area to Cyber

2016: EW School at Ft. Sill becomes subordinate to Cyber School

- Now called "EW College"

<u>1 October 2018: EW officially became part of Cyber Branch</u>

RETIRED

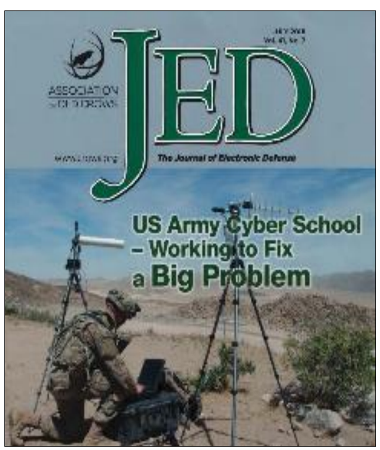
- 17B – EW Officer
- 170B – EW Technician (Warrant Officer)
- 17E – EW Specialist

- 2020/2021 – EW training consolidated at Fort Gordon.

EW's Organizational Journey within the 1945-2019

Association of Old Crows

Original 1964 Logo



EW Professional Organization (1964-Present)

- WWII – Allied Electronic Counter-Measures (ECM) officers code-named "Ravens"
- Post-WWII – EW students started calling their Raven instructors – "Old Crows"
- 1953 – Mel Jackson (US Army Air Forces ECM Pilot in WWII) produces the idea for an "Association of Old Crows" (AOC)
- 1964 – The first AOC banquet and first official due-paying members join
- 1967 – AOC magazine, "Crow Caws" publishes first article; changed to "The Journal of Electronic Defense" (JED) in 1978
- Present Day – AOC has over 14,000 members in 19 countries

CHECK ON LEARNING

Q: What event in American History ultimately led to a dramatic change in Army EW?

A: September 11, 2001 – (9/11)

Q: What was the Dept. of Defense’s response to the RCIED problem?

A: The CREW program - Counter RCIED Electronic Warfare

Q: On what date did EW officially become part of the Cyber Branch?

A: 1 October 2018

Application

- How could you use this information in the future?
- What value does this information have to you?
- What questions do you have for me?

Practical Exercise

- **Objective**: Students will research and analyze real-world applications of Electronic Warfare (EW) in recent conflicts (2020–present), focusing on how EW has shaped battlefield dynamics.
- **Material**: Internet
- **Tasks**: Divide into groups of 5 students and present a one-slider summarizing your findings.
  - How was EW utilized? (Ex: by which side, in which battle)
  - Key outcomes or impacts. (Ex: disruption, countermeasures)
  - Key EW Systems employes (Ex: jammers, drone, DF)

Recent conflicts:

1. Russo-Ukraine
2. Israel-Hamas
3. Armenia-Azerbaijan
4. India-Pakistan

Summary

During this lesson, we discussed:

- Significant Pre-World War I Electronic Warfare Events
- Significant Events for Electronic Warfare during the 20^{th} Century
- Significant Events for Electronic Warfare during the 21^{th} Century
- Emerging EW Technologies
- EW's Organizational Journey within the Army 1945-2019
- Army Cyber Origins Before the Branch and School 1998-2013
- Early History of the U.S. Army Cyber School & Branch 2013-2014
- Various Heraldry Symbols of the Army Cyber School & Branch

Questions?